

Physical Chemistry (II)

Lecture 24

기말고사 안내 및 복습



기말고사 안내

- 시간: 12월 15일(목) 오전 10:30~오후 12:00
- 장소: 화학관 315호(홀수 학번), 114호(짝수 학번)
- 계산기 불필요
- 신분증 지참

시험 범위 및 문제 수

강의 자료 기준 17강~23강

총 다섯 문제(4 + 1)

Overview

Multi-electron systems



Orbital approximation



Accuracy (Lec. 17-18)

- Hartree-Fock method
- Post-HF methods

Concepts (Lec. 19-23)

- Multi-electron atoms
- Molecules

Orbital Approximation

Each electron's spatial part is represented

by an one-electron orbital $\psi(\vec{r})$

- Typically $\psi(\vec{r})$ contains variational parameters
- If $\psi(\vec{r})$ is a linear combination of some functions,

$$\psi(\vec{r}) = c_1\phi_1 + c_2\phi_2 + \cdots + c_K\phi_K$$

- Variational method $\rightarrow$ secular equation (cf. Lec. 20, pp. 11-16)
- Basic concept behind basis sets

Hartree-Fock Method

- The **self-consistent field** (SCF) method
- The atomic wavefunction = **Slater determinant**

For N quantum states $a, b, \dots, n$,

$$\phi_{ab\dots n}(1, 2, \dots, N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

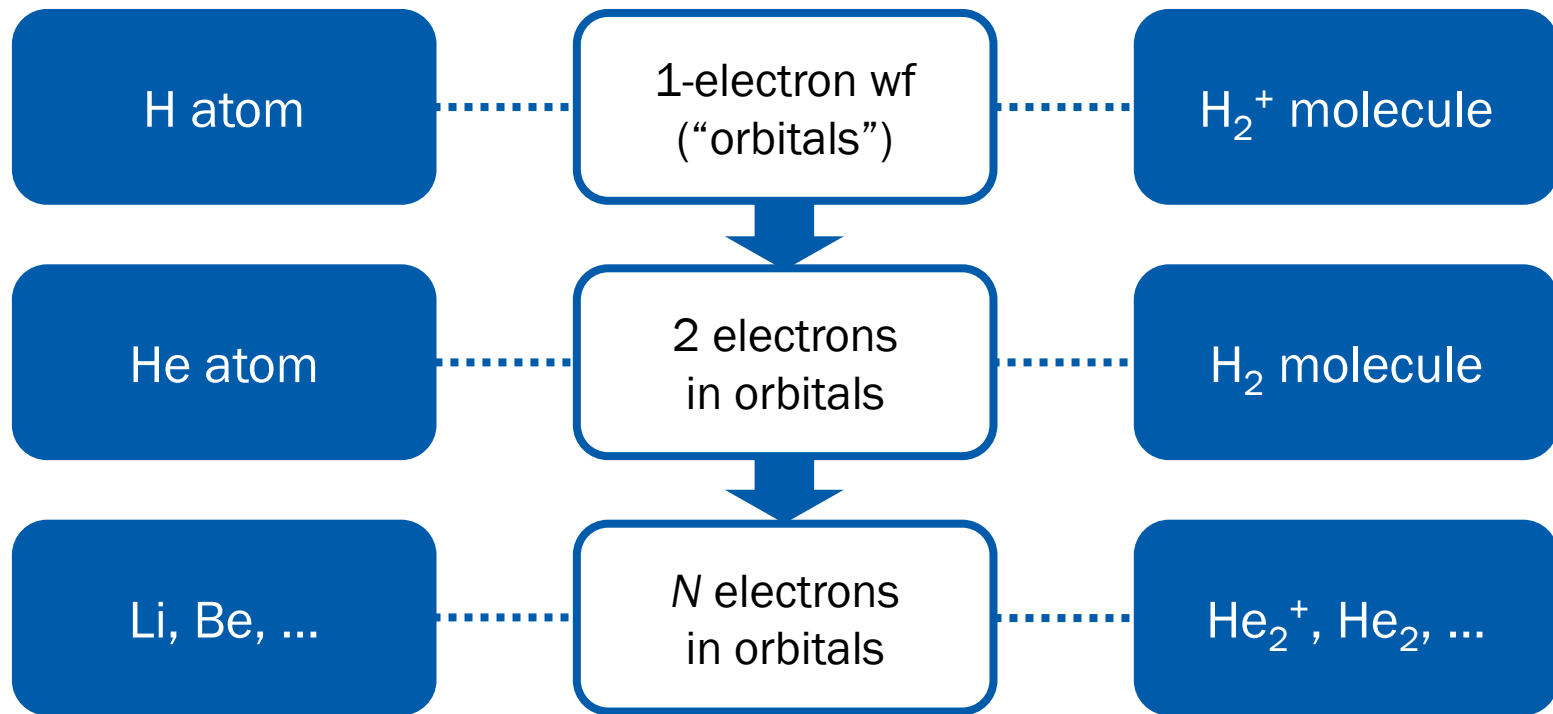
Extension of HF

- Start from one-electron wavefunctions (“orbitals”) u_i

$$\phi = |u_a \quad u_b \quad \cdots \quad u_n|$$

- HF determinants use **occupied orbitals only**
- **Inclusion of unoccupied (virtual) orbitals** may increase accuracy and take care of correlation
- Configuration interaction (CI), coupled-cluster (CC) theory

Multi-Electron Systems



Not necessarily precise

Atoms and Molecules

	Atoms	Diatomic molecules	Polyatomic molecules	Conjugated systems
Construction of orbitals	L19, 13-14	L20	L22 L23, 10-15	L23, 19-25
Electron configurations	L19, 15	<i>Same Aufbau principle</i>		
Electronic states	L19, 16-30	L21, 14-25	L23, 16-18	
Applications	L19, 31-37	L21, 26-32		L23, 26-30

Atoms

- **Orbitals:** based on H-atom wavefunctions
- Electron configurations: Aufbau principle
- **Electronic states** (for light elements): term symbols
- Ground state: Hund's rules
- **Applications**
 - Selection rules → spectroscopy
 - Relativistic effects

Diatomic Molecules

- **Orbitals:** based on atomic orbitals
 - LCAO-MO theory and VB theory
- Electron configurations: Aufbau principle
- **Electronic states:** term symbols
 - Potential energy curves
- Ground state: Hund's rule 1
- **Applications**
 - Molecular vibrations
 - Fate of excited states

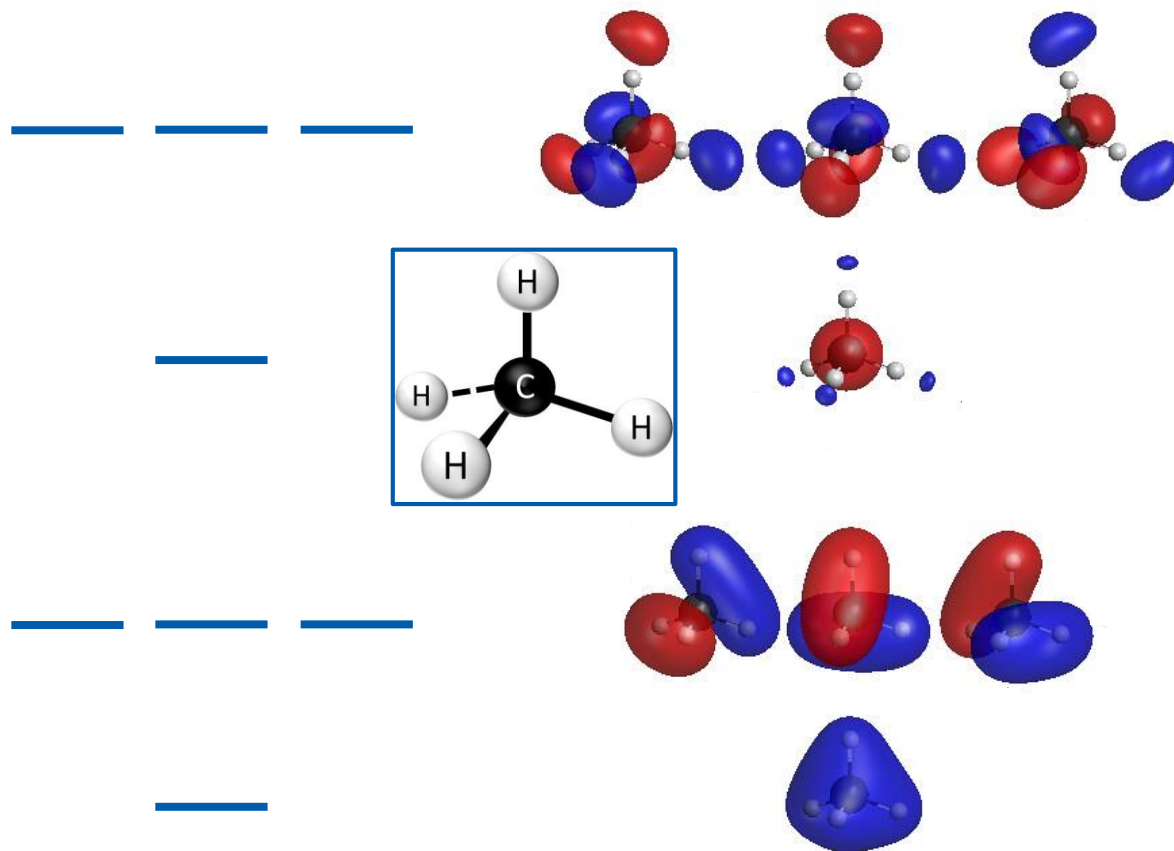
(Small) Polyatomic Molecules

- **Orbitals:** based on atomic orbitals
 - LCAO-MO theory (group theory) and VB theory (hybridization)
- Electron configurations: Aufbau principle
- **Electronic states**
 - Potential energy surfaces

Conjugated Systems

- **Orbitals:** based on π orbitals
 - Hückel MO theory
- Electron configurations: Aufbau principle
- **Applications**
 - UV-visible spectroscopy
 - Aromaticity and antiaromaticity

MOs and Chemical Bonds



(Image: <https://www.flickr.com/photos/69057297@N04/48054283727>)